TEXORA · LECTURE SHEET

Garment Wash Costing

Building the cost of a wash per dozen from its components — chemicals, utilities, labour, machine time, effluent treatment and overhead — and carrying it correctly into the garment's FOB price.

AM-4101 · ADVANCED APPAREL MANUFACTURING-I · TOPIC 5 · SPECIALIZATION — APPAREL MANUFACTURING

1. Why washing is costed separately

A wash is a bought process step, not a rounding error: on a washed garment it can rival the CM charge, and its cost varies sharply with recipe, batch size and liquor ratio. Wash costing therefore follows the same discipline as AM-3207 Topic 6 — build the cost from components, state the basis (per dozen garments), and carry it into the FOB as its own line. The example costed here is the **enzyme + stone wash** of AM-4101 Topic 1, applied to denim garments.

2. The cost build-up per dozen

COMPONENT	WHAT IT CONTAINS	US$ / DOZEN	SHARE
Chemicals & consumables	Desizer, enzyme, softener, neutraliser; pumice consumption share	1.85	41.1%
Labour	Loading, dry-process work share, unloading, batch handling	0.90	20.0%
Utilities	Water, steam for heating baths, electricity for machines and dryers	0.65	14.4%
Machine & depreciation	Washers, hydro-extractor, dryers: depreciation and maintenance per batch-hour	0.45	10.0%
Overhead	Supervision, laboratory and shade-approval work, factory overhead share	0.40	8.9%
Effluent treatment (ETP)	Treatment of discharged bath water to compliance standard	0.25	5.6%

Total wash cost	**4.50 / dozen**	100%

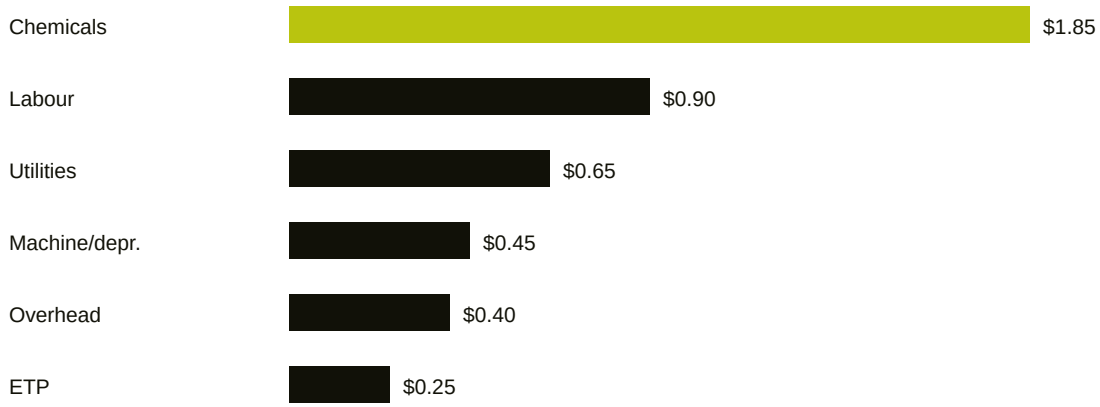


Figure 1 — Wash cost components per dozen (bars to scale). Chemicals dominate — recipe discipline (Topic 1 dosing) is cost discipline.

3. From per-dozen to the order and the FOB

WORKED EXAMPLE — CARRYING THE WASH INTO PRICE

Per piece: 4.50 ÷ 12 = **$0.375 per garment**.

On the course order of 12,000 garments (1,000 dozen): total wash cost = 1,000 × 4.50 = **$4,500**.

In the FOB build-up (AM-3207 Topic 6), the wash enters as its own line: garment cost 4.2714 + 0.375 = 4.6464, and with the 10% margin the washed garment prices at **$5.11 per piece** — not $4.70. Burying the wash inside CM, or forgetting it at quotation and remembering it in production, is how wash orders lose money.

4. What moves the wash cost

- **Liquor ratio.** Bath-acting chemicals, water, steam and ETP all scale with bath volume: cutting LR from 1:10 to 1:8 cuts those components by about a fifth — the strongest single lever, bounded by levelness.
- **Batch size.** Fixed per-batch costs (machine time, handling, lab approval) divide by batch garments; half-empty machines cost nearly double per dozen.
- **Recipe intensity.** Bleach and acid washes add neutralising chemicals and strength-loss risk; stone adds pumice purchase and machine wear; enzyme-only washes are the cheapest fashion wash.
- **Rejection and rewash.** A batch rewashed for shade pays its variable cost twice — first-time shade approval (Topic 1's bulk trial) is a costing control, not only a quality one.

Glossary

TERM	MEANING
Wash cost per dozen	The full cost of washing twelve garments, the standard quotation basis for wash charges.
ETP	Effluent treatment plant: treats process wastewater to the discharge standard; its running cost is allocated to wet processes.
Depreciation per batch-hour	Machine capital cost recovered through a charge for each hour of machine use.
Rewash	Repeating a wash on a failed batch; doubles its variable cost.

Key points and summary

- Wash cost is built from components: chemicals 41%, labour 20%, utilities 14%, machine 10%, overhead 9%, ETP 6% in the worked recipe.
- Enzyme + stone wash: $4.50/dozen = $0.375/piece; $4,500 on the course order.
- The wash is a separate FOB line: washed FOB $5.11 vs unwashed $4.70 per piece.
- Liquor ratio, batch fullness, recipe intensity and rewash rate are the cost levers — all set in Topic 1's process decisions.

Practice

Answer in your own words; brief model answers follow.

1. A simpler enzyme-only wash costs: chemicals $1.20, labour $0.80, utilities $0.55, machine $0.45, overhead $0.40, ETP $0.20 per dozen. Total it, convert to per piece, and cost it for 6,000 garments.
2. At LR 1:8 instead of 1:10, the bath-scaled components (chemicals dosed g/L share, utilities, ETP) fall by 20%. If those components total $2.30 of the $4.50, what is the new wash cost per dozen?
3. Why is a half-loaded washing machine nearly twice as expensive per dozen even though it uses less water?
4. A buyer asks to add a wash after the FOB of $4.70 was quoted. Using this sheet, state the correct new quotation basis.

Answer key

1. Total = 1.20 + 0.80 + 0.55 + 0.45 + 0.40 + 0.20 = $3.60/dozen; per piece = $0.30; 6,000 garments = 500 dozen × 3.60 = $1,800.

2. Saving = 2.30 × 0.20 = $0.46; new cost = 4.50 − 0.46 = $4.04/dozen.
3. Machine time, depreciation, supervision and handling are charged per batch, not per litre: the batch cost is nearly fixed, so halving the garments in it nearly doubles the fixed share carried by each dozen. Water is a minor component.
4. The wash is added as a cost line before margin: (4.2714 + 0.375) × 1.10 = $5.11/pc. Quoting 4.70 + 0.375 = $5.08 would surrender the margin on the wash cost; quoting $4.70 absorbs the whole wash.

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